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Adsorption of hydrogen on Pt(1 1 1) and Pt(1 0 0) surfaces and its role in the HOR

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ABSTRACT

Hydrogen adsorption isotherms, evaluated by combination of cyclic voltammetry and chronoamperometry, are reported on Pt(1 1 1) and Pt(1 0 0) surfaces in 0.1 M $HClO_4$. We found that at $E > 0.05$ V Pt(1 1 1) and Pt(1 0 0) are only partially covered by the adsorbed hydrogen (H_{ad}). On both surfaces, a full monolayer of the adsorbed hydrogen is completed at −0.1 V, i.e. the adsorption of atomic hydrogen is observed in the hydrogen evolution potential region. We also found, that the activity of the hydrogen oxidation reaction is mirrored by the shape of the hydrogen adsorption isotherms, implying that H_{ad} is in fact a spectator in the HOR.

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1. Introduction

The adsorption of hydrogen (dubbed as under potential deposited hydrogen, H_{upd}) on Pt group metals has been over the decades one the main themes in surface electrochemistry; for example see [1–14]. In all of these studies the authors have found that: (i) the formation of H_{upd} adlayer is structure sensitive process; (ii) the Pt(1 1 1)–H_{upd} interaction is rather weak and thus just above the onset for the hydrogen evolution reaction (≈0.05 V) the coverage by H_{upd} on Pt(1 1 1) is only 2/3 of the monolayer (ML); (iii) the interaction of H_{upd} with Pt(1 0 0) is rather strong and on this surface supposedly a full monolayer is formed in the same potential region; and (iv) on both surfaces the monolayer is defined on the bases of one H_{upd} per Pt surface atoms. Although the existing literature provided many important details about the Pt–H_{upd} interaction, one of the key questions still remains open: does the H_{upd} adlayer formation extend in the potential region where the hydrogen evolution reaction (HER) is taking place; and if it does what is a true potential dependent surface coverage by H_{upd} (the so-called adsorption isotherm) on Pt(1 1 1) and Pt(1 0 0). Beyond the question of adsorption isotherms, a central issue in hydrogen electrochemistry concerns the role of H_{upd} on the reaction rates of hydrogen evolution reaction (HER) and the hydrogen oxidation reaction (HOR). For the latter reaction, there are basically two different viewpoints: (i) H_{upd} is indeed an intermediate in the HOR [13,15] and (ii) H_{upd} is a spectator that may block the adsorption of H_2 on active Pt sites [16,17].

This letter provides a detailed electrochemical (cyclic voltammetry, chronoamperometry and rotating disk electrode) analysis of potential dependent hydrogen adsorption on Pt(1 1 1) and Pt(1 0 0) and its the role on the HOR. We present evidence that, in addition to the formation of the H_{upd} adlayer in a "visible" part of cyclic voltammetry (above 0.05 V), an additional adsorption of hydrogen is observed below 0.05 V vs. RHE. To avoid any confusion in the existing terminology for the nature of hydrogen species, we will refer to the adsorbed hydrogen in entire potential region simply as the adsorbed hydrogen (H_{ad}). The effects of adsorbed hydrogen on the HOR are explored by comparing the hydrogen adsorption isotherms and corresponding polarization curves for the HOR.

2. Experimental

The experimental details for crystal preparation and electrochemical measurements have been described elsewhere [18,19]. Chronoamperometry was used to determine the charge associated with the adsorption of atomic hydrogen (Q_{Had}) in two potential regions: $0.05 < E < 0.4$ V, where only H_{ad} (and possibly anion adsorption/desorption) occurs; and $-0.1 < E < 0.05$ V, where H_{ad} and Faradaic (HER) processes are taking place concurrently. As a consequence, two types of responses were expected in each of these two potential ranges. The simulations of the two i vs. t basic shapes, assuming that adsorption process follows the Frumkin isotherm and that the Faradaic reaction is under kinetic control, are shown in Fig. 1a and b.

From Fig. 1a, the current in the pure adsorption/desorption potential region exhibits a sharp current spike, followed by an exponential decay towards zero. The charge (surface coverage) from this relationship can easily be assessed by integration of the I–T

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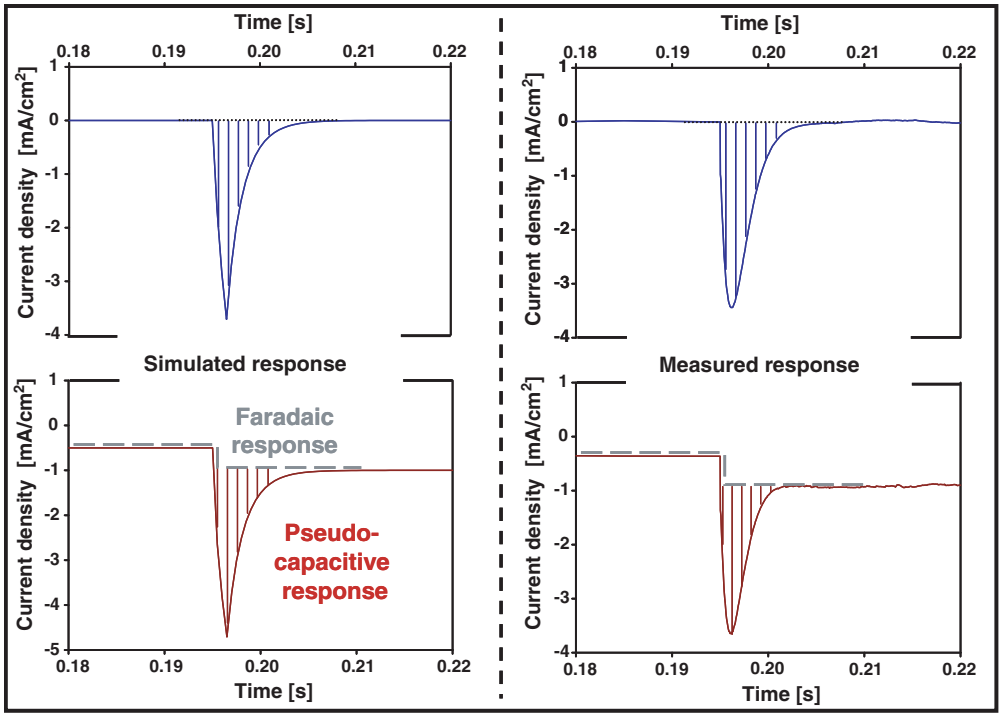


Fig. 1. Chronoamperometric response after a 20 mV potential step: (a) Simulated pseudocapacitive response of a Frumkin type adsorption process; (b) Simulated mixed Faradaic/pseudocapacitive response – Faradaic part of the current is marked as a gray dashed line; (c) A real *I–T* response to a potential step from 0.3 to 0.28 V on Pt(1 1 1) (pure pseudocapacitive response) and (d) A real response to a potential step from 0.04 to 0.02 V (mixed Faradaic/pseudocapacitive response). The charge integration for each case is depicted as the dashed area.

curve. For the potential range within mixed adsorption and Faradic processes (as seen from Fig. 1b), however, the simulated current exhibits a sharp spike, followed by an exponential decay to a constant current value for the HER. The assessment of the charge in this case is not so straightforward and the separation of the Faradaic from pseudocapacitive current requires one important condition to be fulfilled. Namely, the Faradaic reaction has to be under kinetic control, i.e. has to follow Butler–Volmer equation

$$i_K = nFAk^0\left(C_R e^{\alpha_A f E} - C_O e^{-\alpha_C f E}\right) \quad (1)$$

where i_K is the kinetic current, k^0 is the rate constant at $E = 0$, C_R and C_O are the concentrations of the reduced and oxidized species, respectively, α_A and α_C are the transfer coefficients of anodic and cathodic reaction, respectively, and f is simply F/RT. Since there is no time involved in the Eq. (1), the current–time response for such a process is essentially square shaped (see gray dashed line Fig. 1b), making it fairly easy to separate from the pseudocapacitive current. In our potential window, this condition is fulfilled for the HER.

Corresponding chronoamperometric experiments in Fig. 1c and d are in good agreement with simulated curves. In these measurements the electrode potential was set to a value of 0.4 V and held there for 100 ms, long enough to establish steady-state conditions. Then, consecutive 25 potential steps of 20 mV into cathodic direction were made between 0.4 and −0.1 V, waiting 100 ms after each step to reach steady-state conditions. A step of 20 mV was chosen because it gave a reasonable potential resolution as well as signal to noise ratio. When the potential was held in the domain of the HER, rotation of 1600 rpm was also employed to diminish a possible effect of hydrogen bubble formation at the electrode. The experiments were repeated three times for each potential value to give a standard deviation of 8%. To re-construct voltammetric curves of Pt(1 1 1) and Pt(1 0 0) in the hydrogen adsorption potential region, the charge under the curve was integrated (shaded area in Fig. 1) and then transformed (Eq. (2)) into current density that would be obtained at 50 mV/s.

$$i = Q\frac{v}{\Delta E} \quad (2)$$

where i is the current density, Q is the charge density, ΔE is the potential step and v is the sweep rate.

All potentials are given with respect to the reversible hydrogen electrode.

3. Results and discussion

Fig. 2b and c shows typical voltammetric curves for Pt(1 1 1) and Pt(1 0 0) in 0.1 M $HClO_4$. A unique property of the Pt(1 1 1) surface is that the H_{ad} adsorption–desorption region (0.05–0.4 V) is completely separated from the "butterfly" potential region ($E > 0.5$ V), which corresponds to the adsorption of OH/anion [20]. The charge under the curve in the H_{ad} potential region amounts to 160 μC/cm^2. Theoretically, the (1 1 1) surface could accommodate 1.5×10^{15} atoms of H per cm^2, which would correspond to 240 μC/cm^2. The situation is slightly different for the Pt(1 0 0) surface, considering that the H_{ad} adsorption/desorption is coupled with anion/OH desorption/adsorption. To separate charge of H_{ad} from anions the so-called CO displacement method was used [21]. By assuming that no anions are adsorbed at low potentials ($E < 0.1$ RHE), [14] has shown that coverage by adsorbed hydrogen at 0.07 V (starting potential of the voltammetric curve) is close to 208 μC/cm^2. Based on this experiment, it has been concluded that a full hydrogen monolayer is formed on Pt(1 0 0) positive of the onset for the HER (at ca. 0.07 V).

Unfortunately, both cyclic voltammetry and CO displacement technique are essentially useless for determining pseudocapacitance below 0.05 V, where high Faradaic currents of the HER may mask the currents associated with adsorption/desorption processes. As a consequence, it is still puzzling if the H_{ad} formation extends into the potential region where the hydrogen evolution reaction (HER) is taking place on both Pt(1 1 1) and Pt(1 0 0) sur-

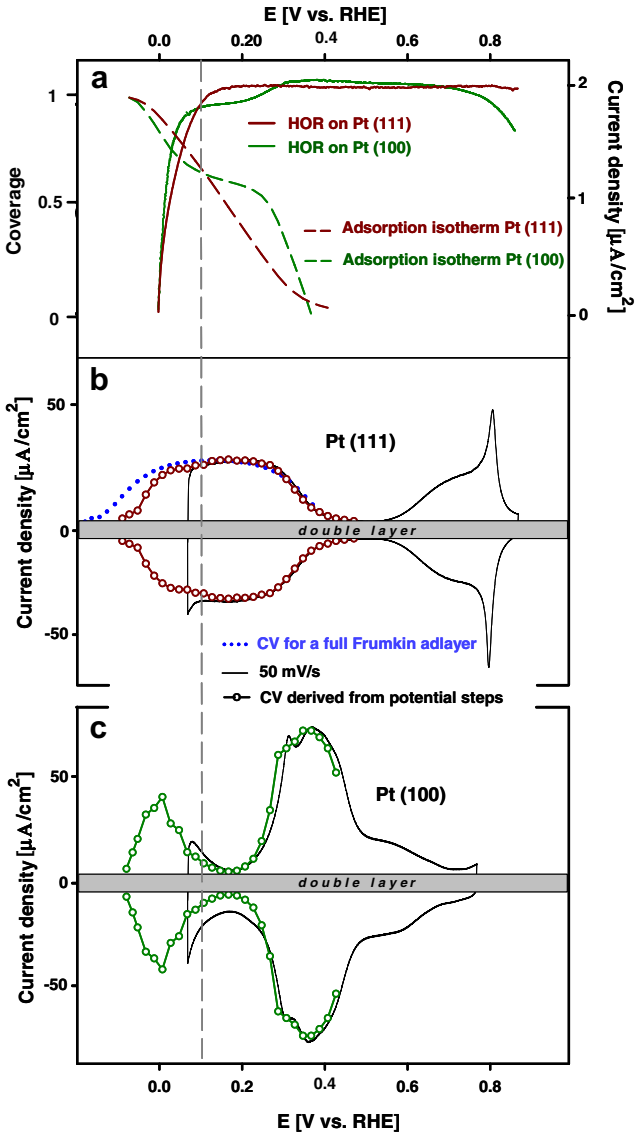


Fig. 2. (a): Adsorption isotherms for Pt(111) – red and Pt(100) – green, obtained by integration of the extended voltammetries. Red and green full lines show the HOR polarization curves on Pt(111) and Pt(100) surfaces, respectively. (b) and (c): Voltammetries for Pt(111) and Pt(100) (black full lines), measured as a response to a classic cyclic potential sweep. Blue dots represent a theoretical voltammetry for a Frumkin adsorption process, with a total charge of 240 μC/cm^2. The green and red curves show the extended voltammograms for Pt(111) and Pt(100), obtained by means of chronoamperometric experiments, described thoroughly in the Section 2.

faces. To determine this, we used *i* vs. time curves (see Section 2). As we discussed above, the results were used first to determine the H_{ad} charge below 0.05 V, then to re-construct voltammograms for Pt(111) and Pt(100) and finally to assess a true hydrogen adsorption isotherms. As shown in Fig. 2b for Pt(111), an additional charge for H_{ad} is observed even below 0.05 V, unambiguously showing that indeed the formation of H_{ad} is not completed in the traditional H_{upd} potential region, as elaborated in theoretical studies (see the most recent and references therein [24]). Furthermore, in both potential regions, a re-constructed CV closely follows an ideal Frumkin isotherm (blue dots in Fig. 2b). The additional charge corresponding to the adsorption of H_{ad} below 0.05 V is (50 ± 4 μC/cm^2), suggesting that a true coverage by adsorbed hydrogen on Pt(111) is significantly larger than 2/3 of a monolayer.

Interestingly, Fig. 2c also unambiguously shows that even on Pt(100) an additional adsorption of hydrogen is observed in the HER potential domain, implying that the previously established adsorption isotherm for the H_{upd} has to be corrected for an additional charge of Q_{Had} = 60 ± 5 μC/cm^2. At this moment it is very difficult to rationalize the existence of this extra charge. The most plausible explanation seems to be that the adsorption of hydrogen and possibly anions below 0.05 V commences at steps and/or other type of defects, which are inherently present on Pt(100) [22,23]. Nonetheless, this result clearly indicates that the existence of H_{ad} below 0.05 V on Pt surfaces is a rather general than unique property of the Pt(111)–H_{ad} interface. Assuming that on both surfaces full coverage is reached when the adsorption process are no longer observed (ca. −0.1 V RHE) the hydrogen adsorption isotherms were obtained by integration of the charge between $-0.1 < E < 0.4$ V. (In this way $\theta = 1$ corresponds to 210 μC/cm^2 of H_{ad} on Pt(111) and to 265 μC/cm^2 of H_{ad} on Pt(100).) In what follows, we demonstrate that the activity for the H_2 oxidation reaction is determined by shapes of hydrogen adsorption isotherms.

From the polarization curves for the HOR on Pt(111) and (100) surfaces, summarized in Fig. 2a, three important pieces of information can be obtained: (i) at potentials below 0.1 V, the (100) surface is more active than the (111) surface; (ii) at potentials above 0.1 V, the (111) is more active than (100) surface and (iii) the (100) exhibits two diffusion limited plateaus, first one starting at 0.07 V and the second one, which corresponds to the theoretical diffusion limited current, starting at 0.3 V. Importantly, the HOR currents are closely related to the surface coverage of adsorbed hydrogen. In particular, hydrogen desorption on Pt(111) is more sluggish at the beginning, so is the HOR. At 0.1 V, however, where the fractional surface coverage by H_{ad} is the same, the activity of the HOR on both surfaces is the same. Also interesting in the present context is that at more positive potentials from 0.1 V, the observed plateau in the HOR at Pt(100) is mirrored by the same one for the hydrogen desorption. Finally at low coverages by H_{ad} the theoretical diffusion currents for the HOR are the same on both surfaces. Clearly, the fact that the approximate concordance observed between the shape of hydrogen adsorption isotherm and the corresponding polarization curve for the HOR suggests that the adsorbed hydrogen is indeed a spectator species which can block the adsorption of H_2. Thus, the structure sensitivity of the HOR is a consequence of structure sensitive adsorption of H_{ad}, which is in harmony with results summarized in Fig. 2 and our latest findings on the relationship between the spectator species and electrocatalytic reactions [19].

4. Conclusion

In conclusion, we found that at $E > 0.05$ V Pt(111) and Pt(100) are only partially covered by the adsorbed hydrogen (H_{ad}). On both surfaces, the process of hydrogen adsorption is completed at −0.1 V, e.g., the formation of H_{ad} is observed in the hydrogen evolution potential region. We also found, that the activity of the hydrogen oxidation reaction is mirrored by the shape of the hydrogen adsorption isotherms, confirming that H_{ad} is in fact a spectator in the HOR.

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